

①

A) V_{th} : terminals a and b are opened

$$\rightarrow 20 + \frac{V_1}{40} + \frac{V_1}{40} + \frac{V_1 - V_2}{10} = 0 \quad i_x = -\frac{V_1}{40}$$

$$800 + 6V_1 - 4V_2 = 0 \Rightarrow 20V_2 = -800$$

$$\boxed{V_2 = -40V}$$

$$\frac{V_2 - V_1}{10} + \frac{V_1}{20} + \frac{V_2 - 40i_x}{20} = 0$$

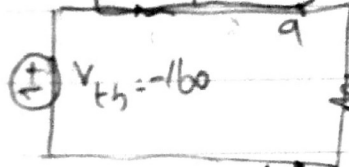
$$4V_2 - V_1 = 0 \Rightarrow V_1 = 4V_2$$

$$\boxed{V_1 = -160V}$$

②

$$V_{th} = V_{ab} = V_1 = \boxed{-160V}$$

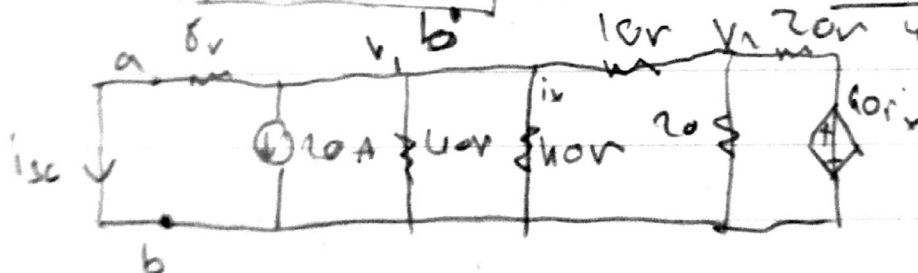
$$R_{th} = 16\Omega$$



$$R = R_{th} = 16\Omega$$

$$b = P_{max} = \frac{V_{th}^2}{4R}$$

$$\frac{160 \cdot 160}{4 \cdot 16} = 400W$$



$$i_{sc} = \frac{V_1}{8} \quad i_x = -\frac{V_1}{40}$$

$$\frac{V_1}{8} + 20 + \frac{V_1}{40} + \frac{V_1}{40} + \frac{V_1 - V_2}{10} = 0$$

$$\frac{V_2 - V_1}{10} + \frac{V_2}{20} + \frac{V_2 - 40i_x}{20} = 0$$

$$V_1 = 4V_2$$

$$11V_1 - 4V_2 + 800 = 0$$

$$44V_2 - 4V_2 = -800$$

$$V_2 = -20V \quad V_1 = 4V_2 = -80V$$

$$i_{sc} = \frac{V_1}{8} = -10A$$

$$R_{th} = \frac{V_{th}}{i_{sc}} = \frac{-160V}{-10A}$$

$$= \boxed{16\Omega}$$

①

$$A) v_1 < 1 \frac{1 \cdot k\Omega}{1k\Omega + 9k\Omega} < 0.1V$$

$$\frac{v_1 - v_2}{10k\Omega} + \frac{v_1 - v_0}{100k\Omega} = 0$$

$$\rightarrow v_0 = v_1 + 10(v_1 - v_2) = 11v_1 - 10v_2 = 11 \cdot 0.1V - 10V_A$$

$$b) -10 \leq v_0 \leq 15$$

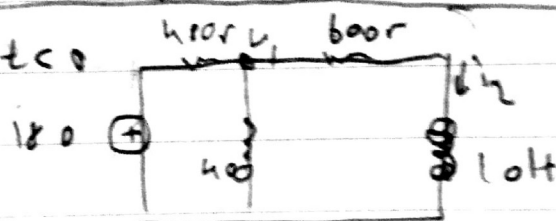
$$-10 \leq 1.1 - 10v_2 \leq 15$$

$$-11.1 \leq -10v_2 \leq 13.9$$

$$-1.39V \leq v_2 \leq 1.11V$$

③ $t < 0, t > 0$

Ⓐ



$$\frac{v_1 - 180}{400} + \frac{v_1}{400} + \frac{v_1}{600} = 0 \quad \frac{v_1 - 180}{2k} + \frac{v_1}{200} + \frac{v_1}{300} = 0$$

$$1.5v_1 = 5 \cdot 180 \quad v_1 = 60V \rightarrow i_L(0) = \frac{v_1}{600} = 0.1A$$

$$t < 0, t > 0 \quad \begin{aligned} & \text{Circuit diagram for } t < 0: \text{ A loop with a 30V source, a 400Ω resistor, and a 600Ω resistor in series. The voltage across the 600Ω resistor is } v_1. \\ & \text{Circuit diagram for } t > 0: \text{ A loop with a 30V source, a 1000Ω resistor, and a 10H inductor in series. The voltage across the inductor is } v_L. \\ & i_L(t) = i_L(0) e^{-t/\tau} + 30 = 0.1 e^{-\frac{1000t}{10}} + 30 \\ & \boxed{= 0.1 e^{-100t} + 30} \end{aligned}$$

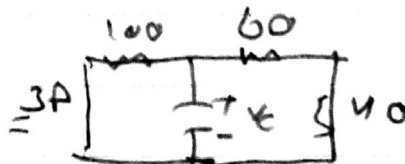
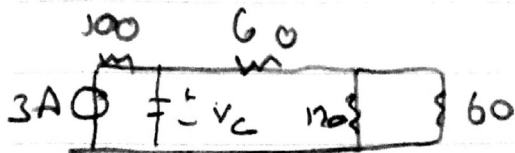
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$$\begin{aligned} v_L(t) &= L \frac{di_L(t)}{dt} = 10 \frac{d}{dt} (0.1 e^{-100t}) + 30 \\ &= -100 e^{-100t} + 30 \end{aligned}$$

④

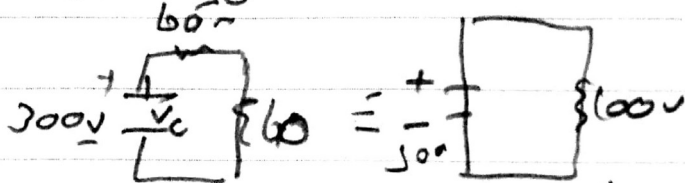
A

for $t < 0$



$$V_c(0) = 3 \cdot 100 = 300 \text{ V}$$

for $t \geq 0$

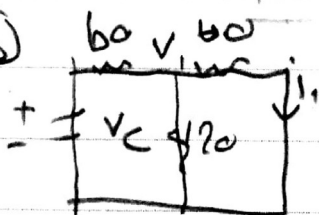


$$V_c(t) = V_c(0) \cdot e^{-\frac{t}{\tau_{eqc}}}$$

$$V_c = 300 e^{-\frac{t}{100 \cdot 200 \cdot 10^{-6}}} \quad t \geq 0$$

$$= 300 e^{-50t} \quad t \geq 0$$

B



$$\frac{V_1 - V_c}{60} + \frac{V_1}{120} + \frac{V_1}{60} = 0$$

$$5V_1 - 7V_c = 0$$

$$\Rightarrow V_1 = \frac{7}{5} V_c = 0.4 \cdot V_c \quad 120 \cdot e^{-50t} \quad t \geq 0$$

$$i_1 = \frac{V_1}{60} = 7 e^{-50t} \quad t \geq 0$$